The University of Texas at Austin
In-Class Work 17

M378K Introduction to Mathematical Statistics

November 06, 2026

Relative Efficiency.

Definition 17.1: *Given two unbiased estimators $\hat{\theta}_1$ and $\hat{\theta}_2$, the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is defined as*

$$\text{eff}\left(\hat{\theta}_1, \hat{\theta}_2\right) = \frac{\text{Var}\left[\hat{\theta}_2\right]}{\text{Var}\left[\hat{\theta}_1\right]}.$$

Problem 17.2: *Let Y_1, Y_2 be a random sample from the exponential distribution with the unknown parameter θ.*

i. The estimator $\hat{\theta}_1 = (Y_1 + Y_2)/2$ for θ is proposed. What is its variance?
ii. The estimator $\hat{\theta}_2 = cY_{(1)}$ for θ is proposed. Find the constant c such that $\hat{\theta}_2$ is an unbiased estimator of θ. What is its variance?
iii. Calculate the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$.

Solution:

i.
$$\text{Var}\left[\hat{\theta}_1\right] = \text{Var}\left[\frac{Y_1 + Y_2}{2}\right] = \frac{1}{4}(\text{Var}\left[Y_1\right] + \text{Var}\left[Y_2\right]) = \frac{2\tau^2}{4} = \frac{\tau^2}{2}.$$

ii. By definition, $Y_{(1)} = \min(Y_1, Y_2)$. As we have proved earlier in the class

$$Y_{(1)} \sim E\left(\frac{\tau}{2}\right)$$

So, $\mathbb{E}\left[Y_{(1)}\right] = \frac{\tau}{2}$. In order to have $\hat{\theta}_2$ be unbiased, we must set $c = 2$. Since, $\text{Var}\left[Y_{(1)}\right] = \frac{\tau^2}{4}$, we have that

$$\text{Var}\left[\hat{\theta}_2\right] = \text{Var}\left[2Y_{(1)}\right] = 4\left(\frac{\tau^2}{4}\right) = \tau^2.$$

iii.
$$\text{eff}\left(\hat{\theta}_1, \hat{\theta}_2\right) = \frac{\text{Var}\left[\hat{\theta}_2\right]}{\text{Var}\left[\hat{\theta}_1\right]} = \frac{\frac{\tau^2}{\tau^2}}{2} = 2.$$